



Lecture 4:

How do we build electronics?

ECE 340 – Electronic Circuits I Sec. 1

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09/15/2026

(textbook sections 3.1-3.4)



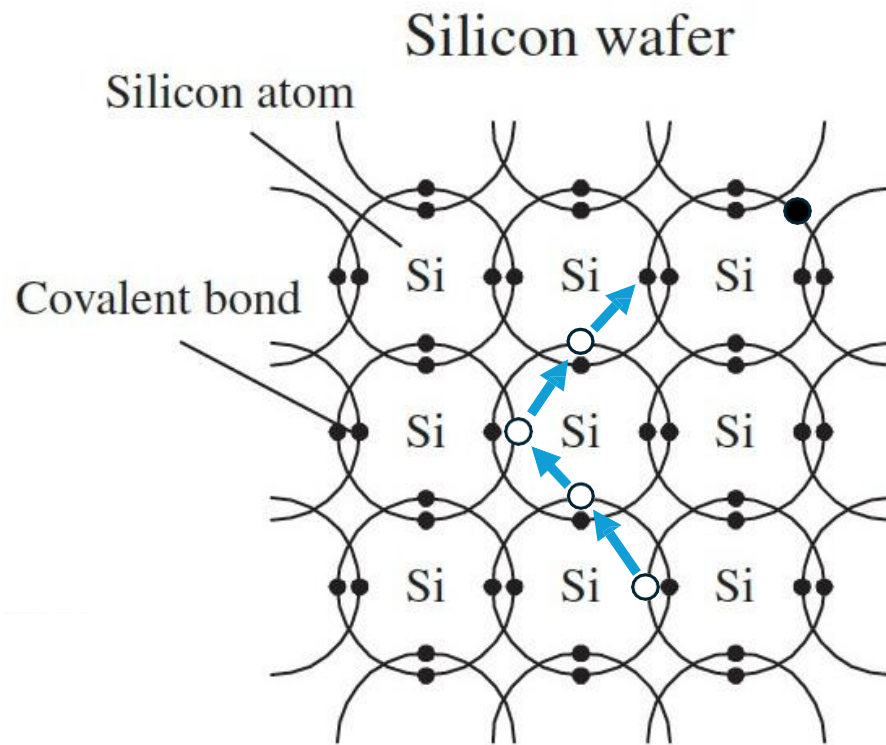
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Today's Outline

1. Intrinsic/Extrinsic semiconductors
2. Electron and hole movement
3. Drift current
4. Diffusion current



Si in bulk



A silicon wafer in pure form doesn't contain any free charge carriers; all the electrons are locked up into covalent bonds between neighboring atoms.

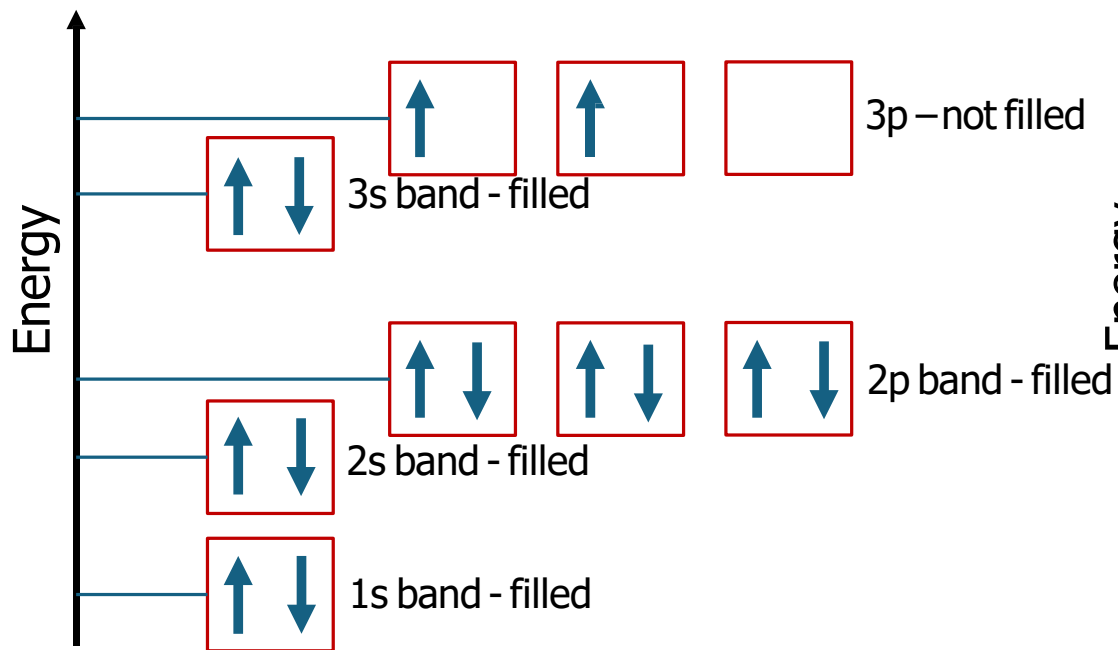
(At zero degrees Kelvin, they are locked up completely; at higher temps, they may jump out and jump back in)

- or they may jump out, leaving what is called a "hole", which is a positive "charge carrier"
- other electrons will jump into the hole, allowing the positive charge to move around.

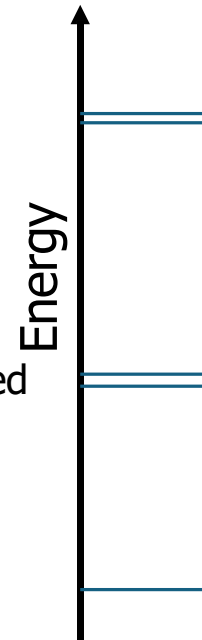
But still, net charge is zero; we don't have a useful device with just plain Si...

Energy band diagram

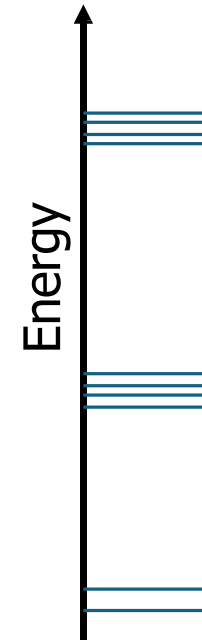
Energy is quantized for few atoms



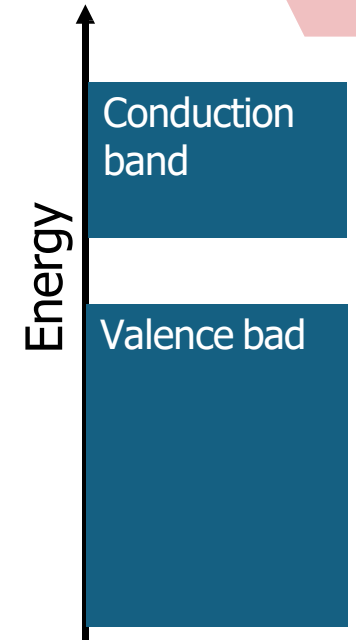
1 Si atom



1 Si atom



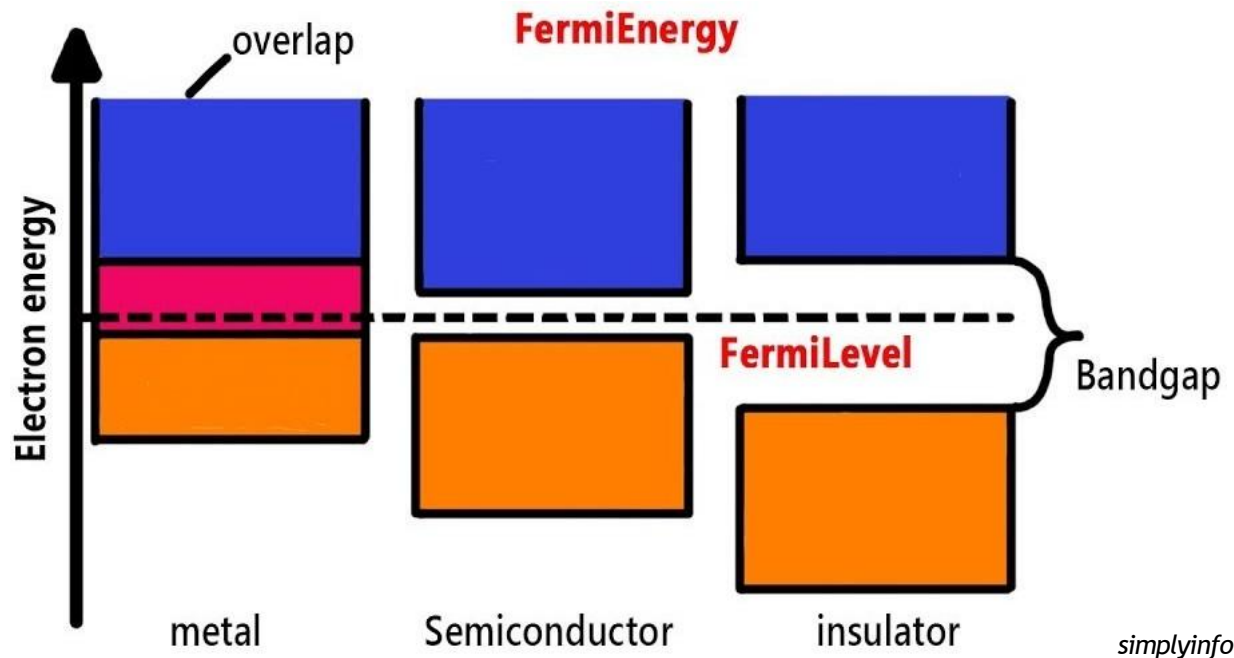
2 Si atoms



Si crystal

Metals vs. semiconductors vs. insulators

Band gap for Si ~ 1.12 eV
Band gap for Ge ~ 0.7 eV
Band gap for GaAs ~ 1.4 eV



overlapping
bands → regular
conduction

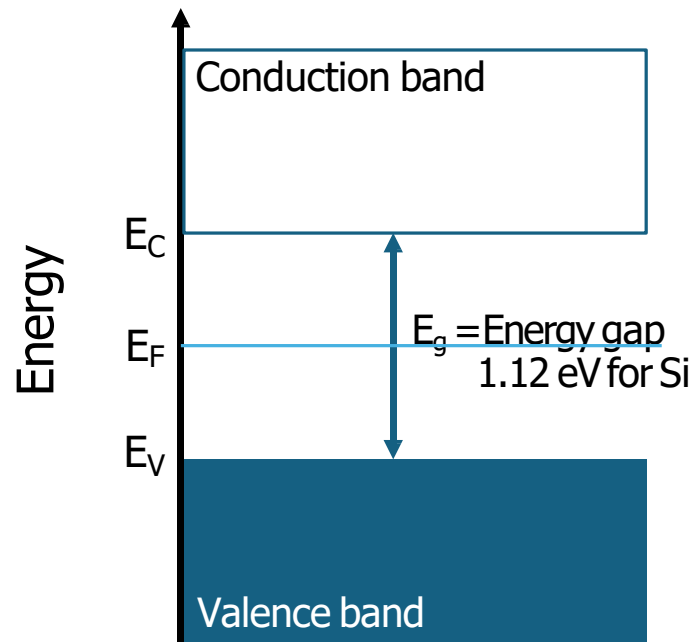
“the goldilocks
gap” → controlled
conduction

typical band gaps for
semiconductors are
between 0-3 eV

widely separated
bands → no
conduction

typical band gaps for
insulators are > 5 eV
e.g., SiO₂ E_g = GeV

Band diagrams – more detail

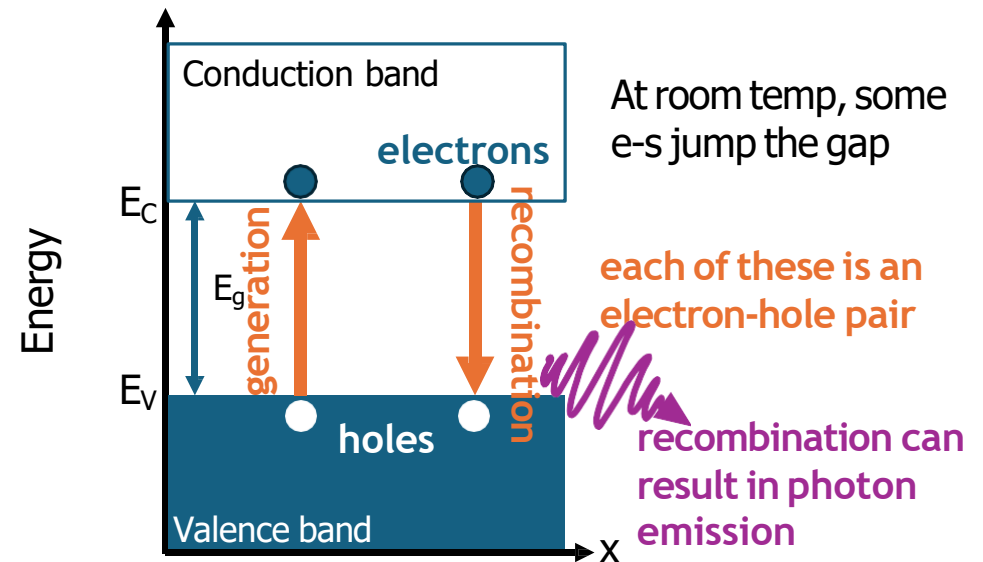


intrinsic Si crystal

1 eV = 1.6×10^{-19} J, the energy acquired when an electron (e^-) falls thru a potential of 1 V

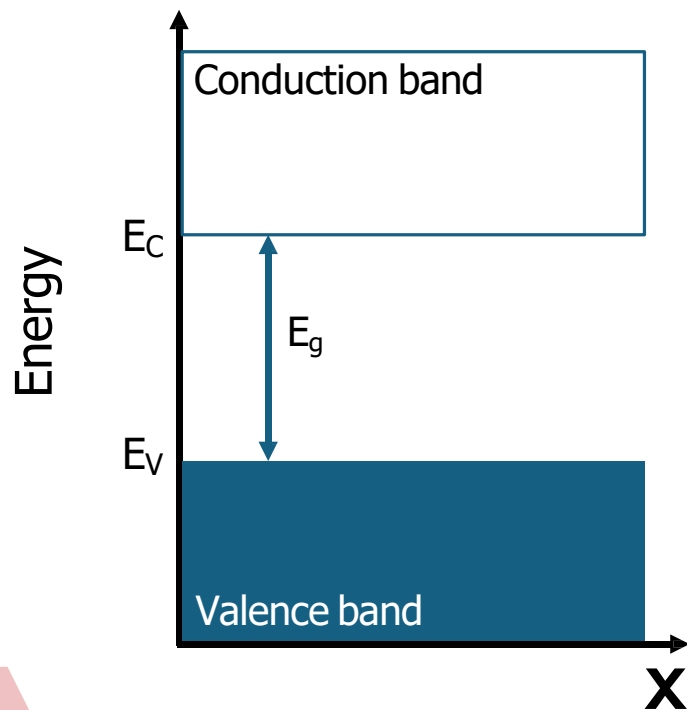
No electron states in the gap! But because they tunnel from valence to conduction and back, there is a nonzero probability of finding electrons there.

At thermodynamic equilibrium, the **Fermi energy** level has a 50% probability of being occupied at any given time.

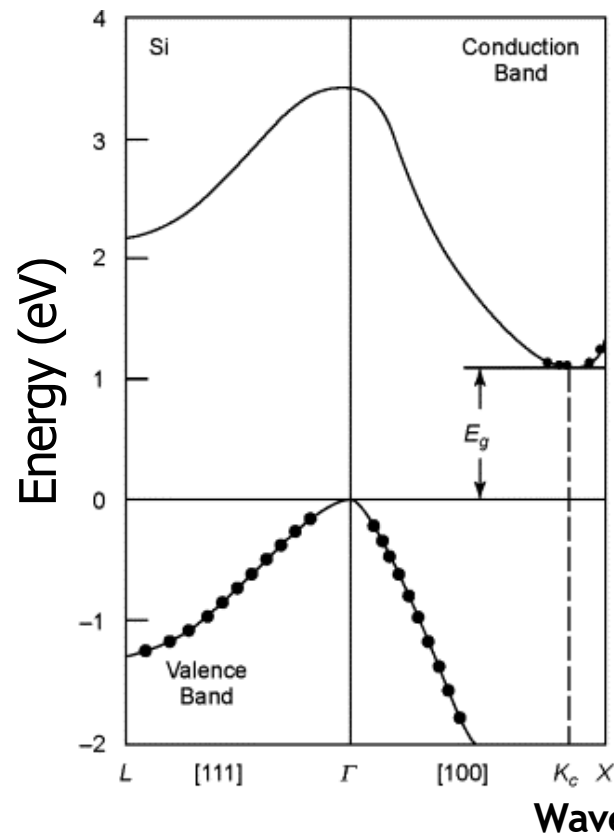


E-k diagrams – indirect band gap

Used for materials characterization, based on momentum of free electrons within the material. Details are beyond the scope of this course but check out the references below for more!



(Our band diagram)



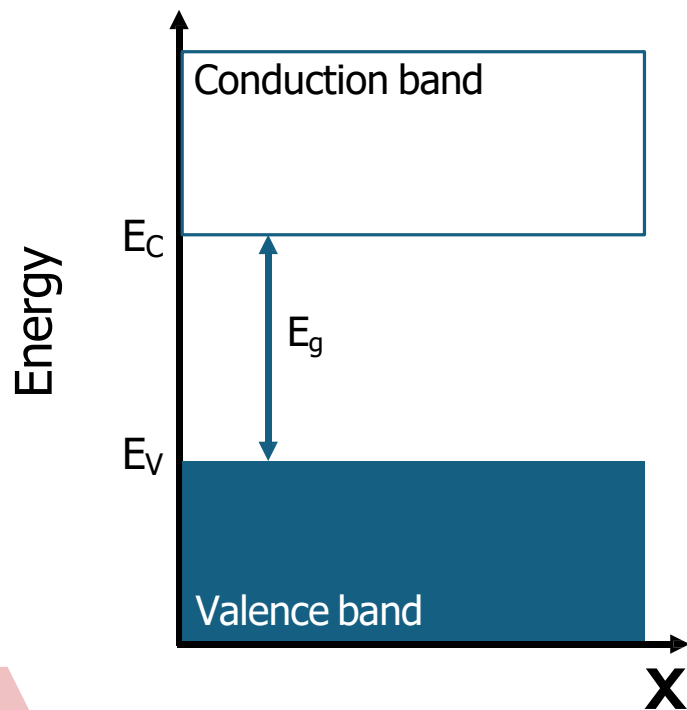
At 0 K the conduction band is empty! At room temp, there is spurious generation C recombination

k - https://mrlweb.mrl.ucsb.edu/~seshadri/2021_218/Bands-k-space.pdf

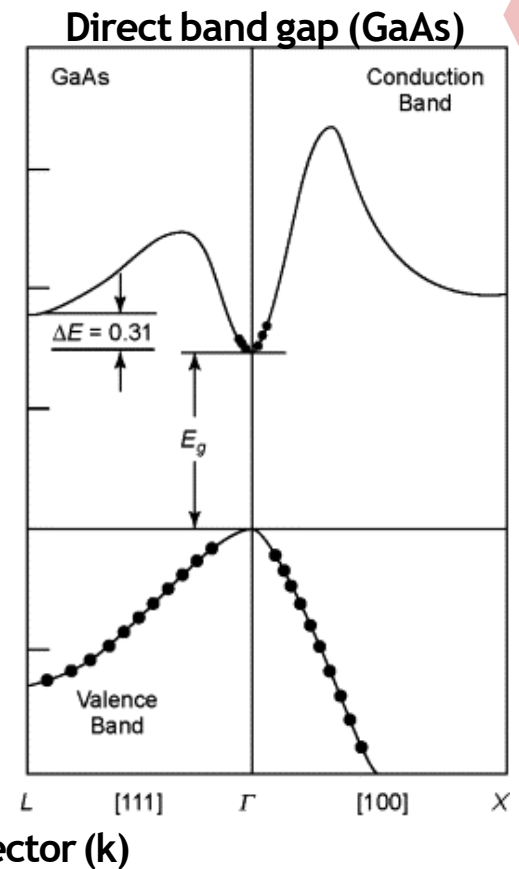
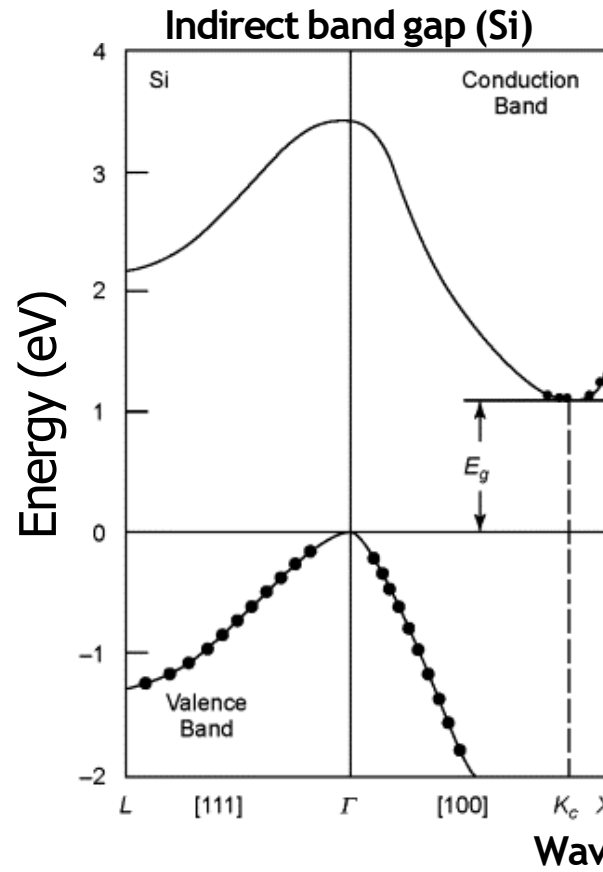
<https://www.sciencedirect.com/topics/engineering/energy-band-theory>

E-k diagrams – direct band gap

Used for materials characterization, based on momentum of free electrons within the material. Details are beyond the scope of this course but check out the references below for more!



(Our band diagram)



<https://www.sciencedirect.com/topics/engineering/energy-band-theory>

Spicing it up

Just like we add spices to our soups, we add dopants to our silicon

*In doing so, we **purposely** make our Si impure!*

Adds negative charge
n-type dopants

+Si = n-type Si

Adds positive charge
p-type dopants

+Si = p-type Si

			Adds positive charge										<i>n</i> -type dopants +Si = <i>n</i> -type Si							
Nonmetals	1																	2		
	H																	He		
Metals	2	3	4											5	6	7	8	9	10	
		Li	Be											B	C	N	O	F	Ne	
	3	11	12	Transition metals (sometimes excluding group 12)										13	14	15	16	17	18	
		Na	Mg											Al	Si	P	S	Cl	Ar	
	4	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	
		K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr	
	5	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	
		Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe	
	6	55	56		71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
		Cs	Ba	La to Yb	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
	7	87	88		103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118
		Fr	Ra	Ac to No	Lr	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og
		s-block (plus He)		f-block	d-block										p-block (excluding He)					

Lanthanides

57	58	59	60	61
La	Ce	Pr	Nd	Pm

Actinides

89	90	91	92	93
Ac	Th	Pa	U	Np

Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No
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Example: compare # of valence electrons (e.g., 5 for As) with the number of electrons of the atom it replaces (e.g., 4 for Si)

Nomenclature

Doping = purposely introducing impurities into the crystal

Intrinsic semiconductor = pure, undoped

Extrinsic semiconductor = doped

2 types of dopants:

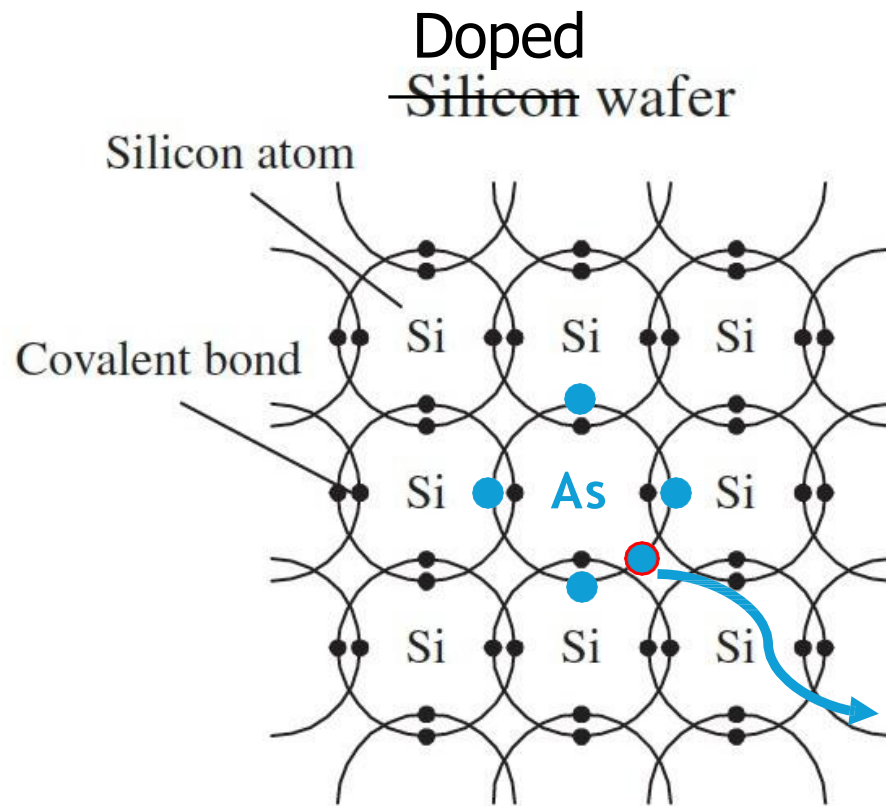
1. One that adds positive charge = *acceptor*
[of electrons]
2. One that adds negative charge = *donor*
[of electrons]

2 types of doped semiconductors:

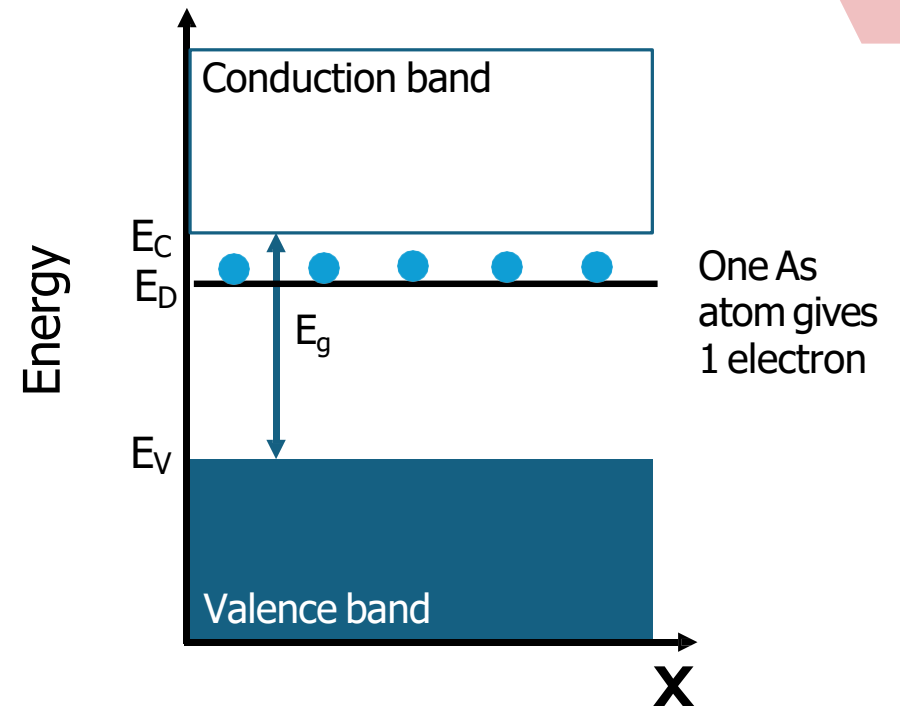
1. *p-type* (doped with acceptors)
2. *n-type* (doped with donors)

Acceptor	Intrinsic	Donor
5 B	6 C	7 N
13 Al	14 Si	15 P
31 Ga	32 Ge	33 As
49 In	50 Sn	51 Sb

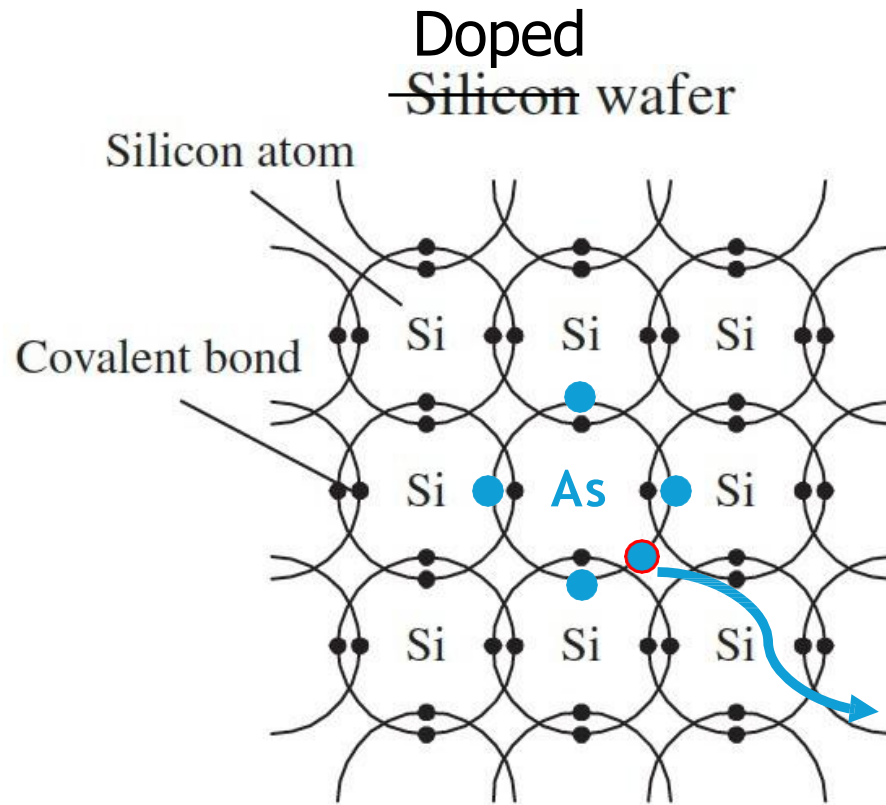
How do dopants change the lattice?



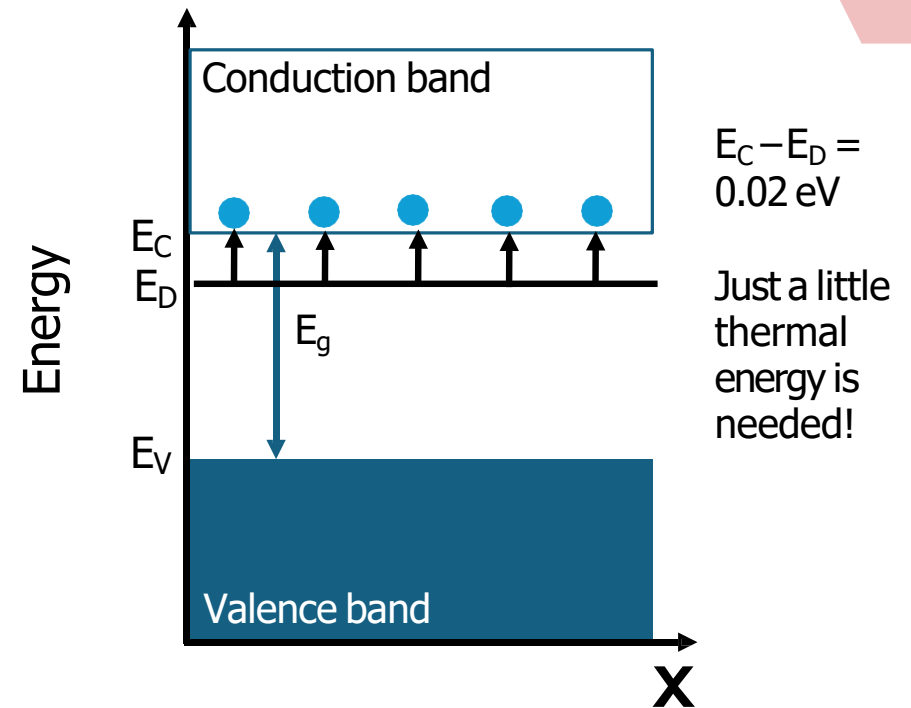
n-type (Donors)



How do dopants change the lattice?



n-type (Donors)



How do dopants change the lattice?

n-type (Donors)

Typical concentration, N_D , is $\sim 10^{15} - 10^{21}$ per cm^3

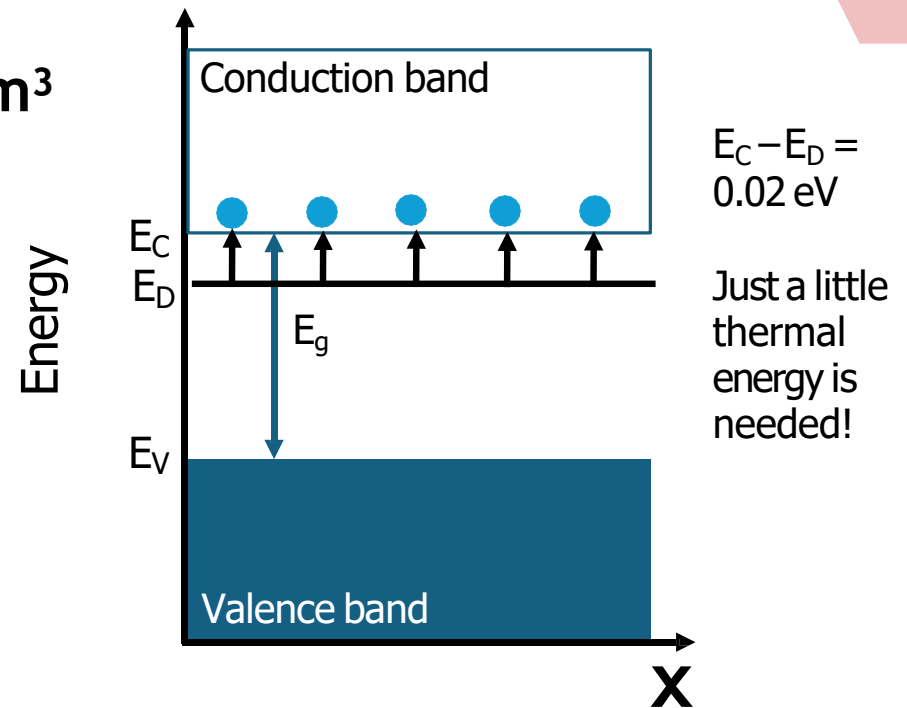
$n_n = N_D$ (donors are slightly positive)

$$n_i^2 = p_n n_n$$

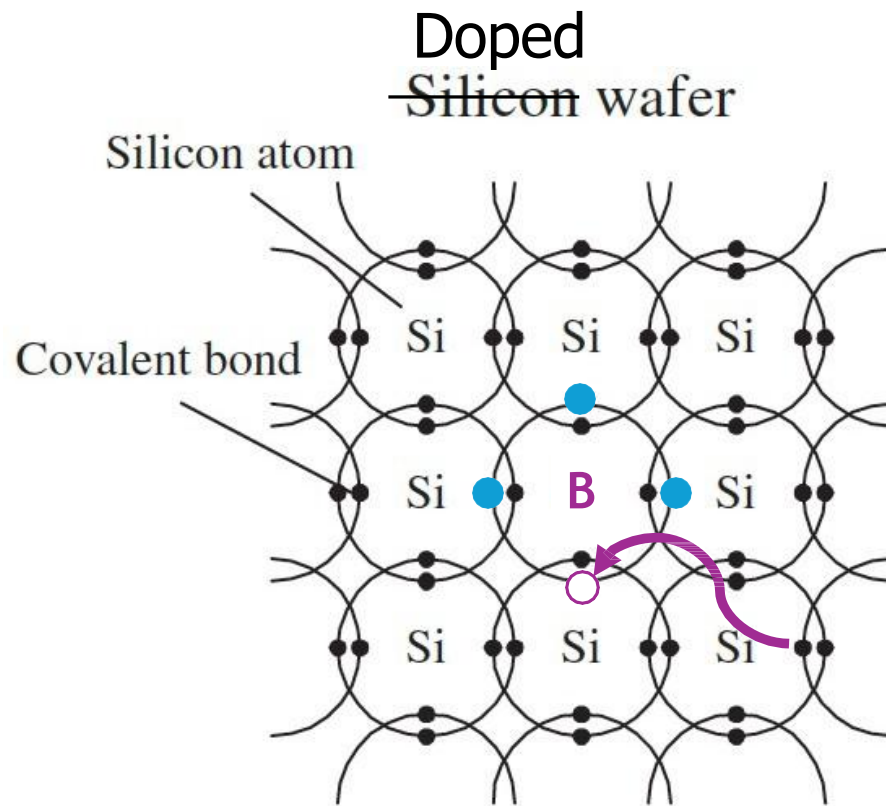
$$\frac{n_i^2}{N_D} = p_n$$

$p_n \leq 10^5$ per cm^3 (typically)

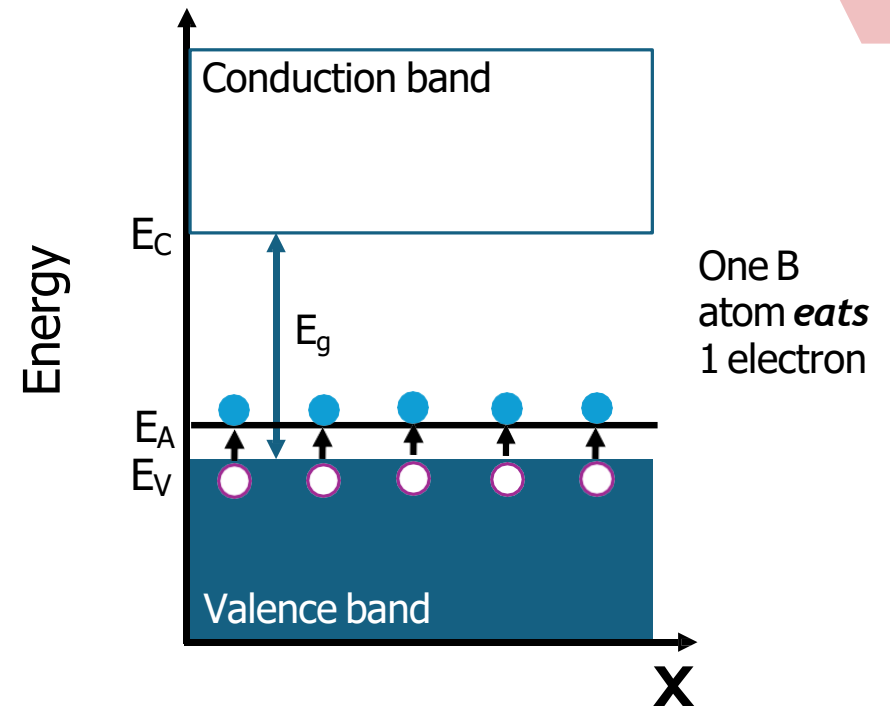
(Majority carriers in n-type are electrons)



How do dopants change the lattice?



p-type (Acceptors)



How do dopants change the lattice?

p-type (Acceptors)

Typical concentration, N_A , is $\sim 10^{16} - 10^{18}$ per cm^3

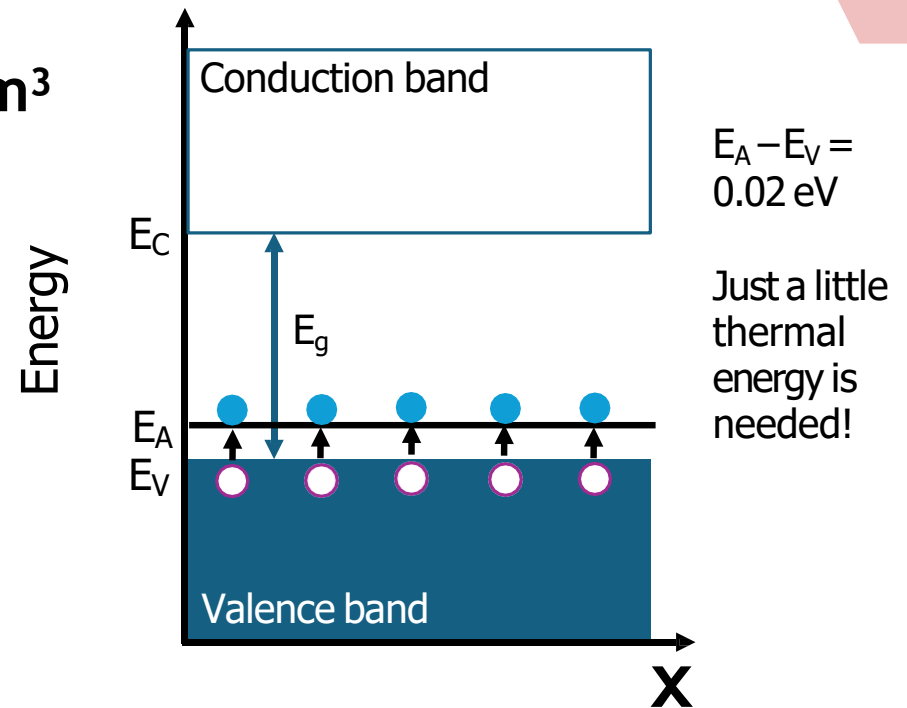
$p_p = N_A$ (acceptors are slightly negative)

$$n_i^2 = p_p n_p$$

$$\frac{n_i^2}{N_A} = n_p$$

$n_p \leq 10^6$ per cm^3 (typically)

(Majority carriers in p-type are holes)



Intrinsic vs. extrinsic carrier concentration

Intrinsic (no dopants)

n = concentration of free electrons

p = concentration of holes

$$n = p = n_i$$

where n_i is the number of free electrons and holes in a unit volume (cm^3)

How we typically express carrier concentration:

$$pn = n_i^2$$

For Si at room temp,

$$n_i \approx 1.5 \times 10^{10} / \text{cm}^3$$

Extrinsic (doped Si)

n-type

Typical donor concentration,

N_D , is $\sim 10^{15} - 10^{21}$ per cm^3

$$\frac{n_i^2}{N_D} = p_n$$

p-type

Typical acceptor concentration,

N_A , is $\sim 10^{16} - 10^{18}$ per cm^3

$$\frac{n_i^2}{N_A} = n_p$$

Carrier concentration in
intrinsic silicon (cm^{-3})

$$n_i = BT^{3/2} e^{-E_g/2kT}$$

$$B = 7.3 \times 10^{15} \text{ cm}^{-3} \text{ K}^{-3/2}$$

$$E_g = 1.12 \text{ eV}$$

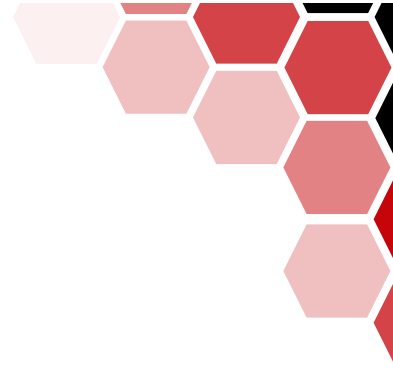
$$k = 8.62 \times 10^{-5} \text{ eV/K}$$

$$n_i = 1.5 \times 10^{10} / \text{cm}^3$$

Example 1(300 K)

What are the electron and hole concentrations in arsenic-doped Si ($N_D = 10^{17} \text{ /cm}^3$) at 300 K?

We can use our previous result $1.5 \times 10^{10} \text{ /cm}^3$



Example 2

For a silicon crystal doped with boron, what must N_A be if at $T=300$ K the electron concentration drops below the intrinsic level by a factor of 10^6 ?

$$n_i = 1.5 \times 10^{10} / \text{cm}^3$$

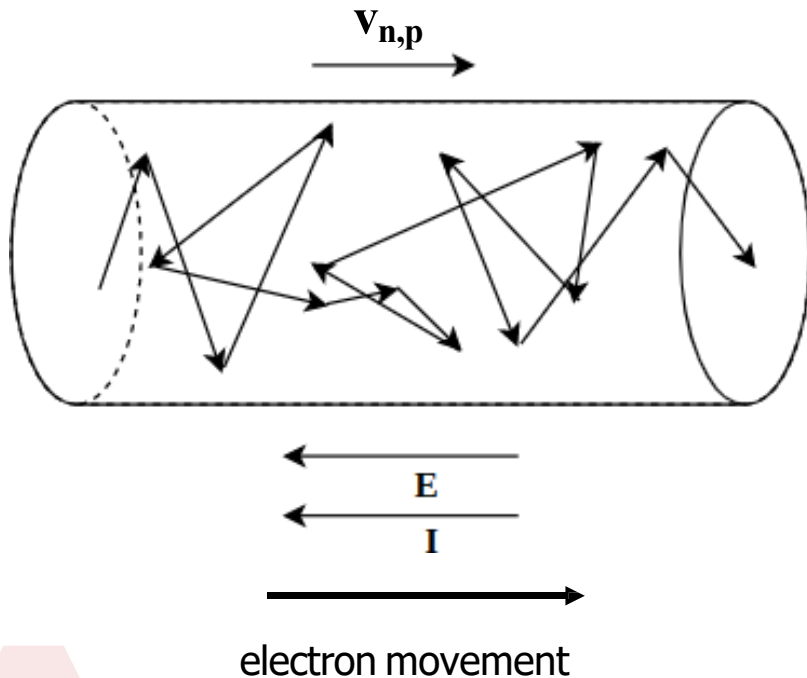


Current type 1: Drift current

Turning ON an electric field

Drift current

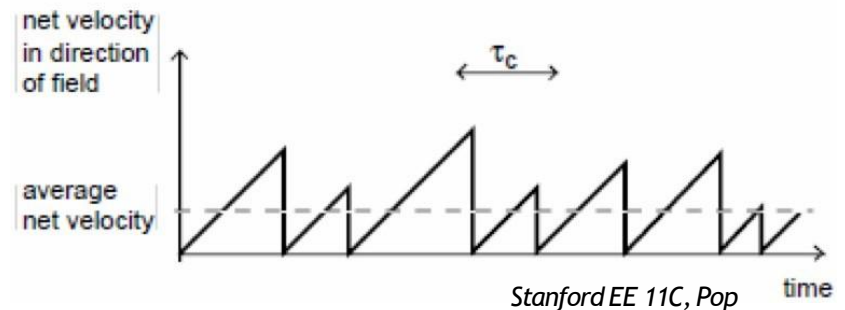
Charges move due to Electric Field



The force on an individual charge

- $F = \pm qE$
- $F = m^*a \rightarrow a = \pm qE/m_{n,p}^*$
- Between collisions, carriers accelerate along E field:
 - $v_n(t) = a_n t = -qEt/m_n$ for electrons
 - $v_p(t) = a_p t = +qEt/m_p$ for holes

But when they bump into the lattice, they slow down again!
(every τ_c collision time, velocity is random)

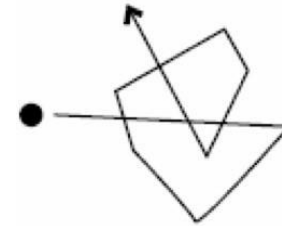


What can we do with doped Si?

On its own? NOTHING!

Without an electric field, we just have boring materials.

- The secret life of C-band electrons (or V-band holes): they are essentially free to move around, how?



The collision time (with what?) is of the order $\tau_C \sim 0.1$ ps
Instantaneous velocity given by thermal energy: $v_T \sim 10^5$ m/s
So the average distance (mean free path) traveled between scattering events?

$$L \sim v_T \tau_C$$

But the total distance traveled averages out to zero!

Turning ON an electric field

Current flow type 1: Drift

Charges move due to Electric Field

Avg. velocity in E-field:

$$|\vec{v}_{n,p}| = \frac{qE\tau_C}{m^*}$$

v-to-E proportionality constant is the **carrier mobility**: $|\vec{v}_{n,p}| = \mu_{n,p}\vec{E}$

$$\mu_{n,p} = \frac{q\tau_C}{m_{n,p}^*} \quad \ll \text{This is very important!}$$

For electrons: $v_{n,drift} = -\mu_n E$

For holes: $v_{p,drift} = \mu_p E$

The force on an individual charge

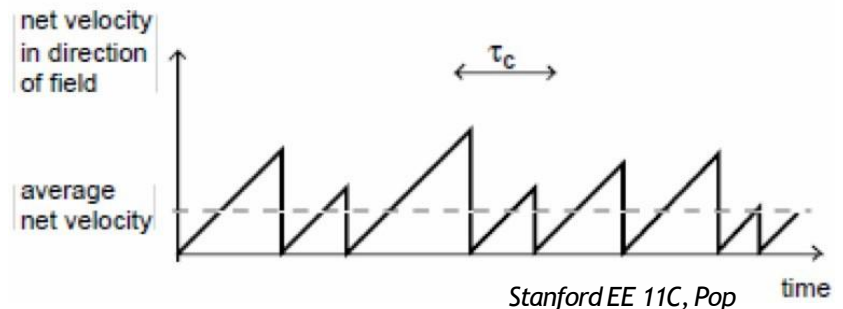
- $F = \pm qE$

- $F = m^*a \rightarrow a = \pm qE/m_{n,p}^*$

Between collisions, carriers accelerate along E field:

- $v_n(t) = a_n t = -qEt/m_n$ for electrons
- $v_p(t) = a_p t = +qEt/m_p$ for holes

but when they bump into the lattice, they slow down again!
(every τ_C collision time, velocity is random)



Mobility

$$\mu_{n,p} = \frac{q\tau_c}{m_{n,p}^*}$$

Mobility is a measure of how easily carriers **drift** in E-field

As m decreases, (particle is lighter), μ ...

As τ_c increases, (longer time b/w collisions), μ ...

	Si	Ge	GaAs	InAs	Graphene / CNTs
μ_n (cm ² /V·s)	1400	3900	8500	30000	40000
μ_p (cm ² /V·s)	470	1900	400	500	40000

electrons are more mobile!

Current density:
$$\vec{J}_n = -q\vec{n}\vec{v}_{n,drift} = qn\mu_n\vec{E} \left[\frac{A}{cm^2} \right]$$

$$\vec{J}_p = qp\vec{v}_{p,drift} = qp\mu_p\vec{E} \left[\frac{A}{cm^2} \right]$$

$$\vec{J} = \vec{J}_n + \vec{J}_p = q(n\mu_n + p\mu_p)\vec{E}$$

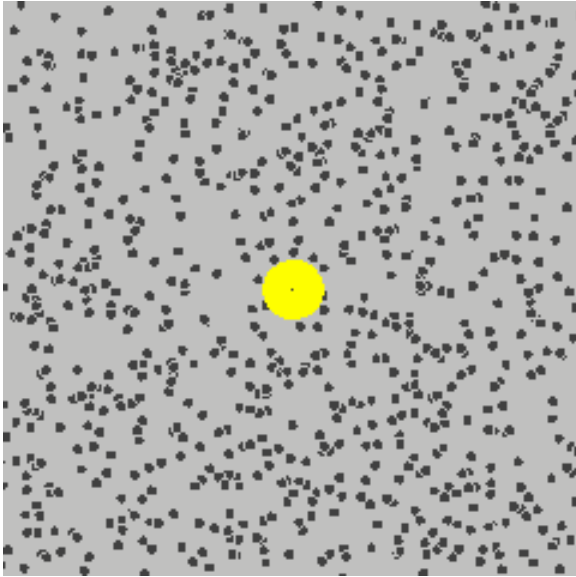
Conductivity, σ

Units are in cm²/ V·s :

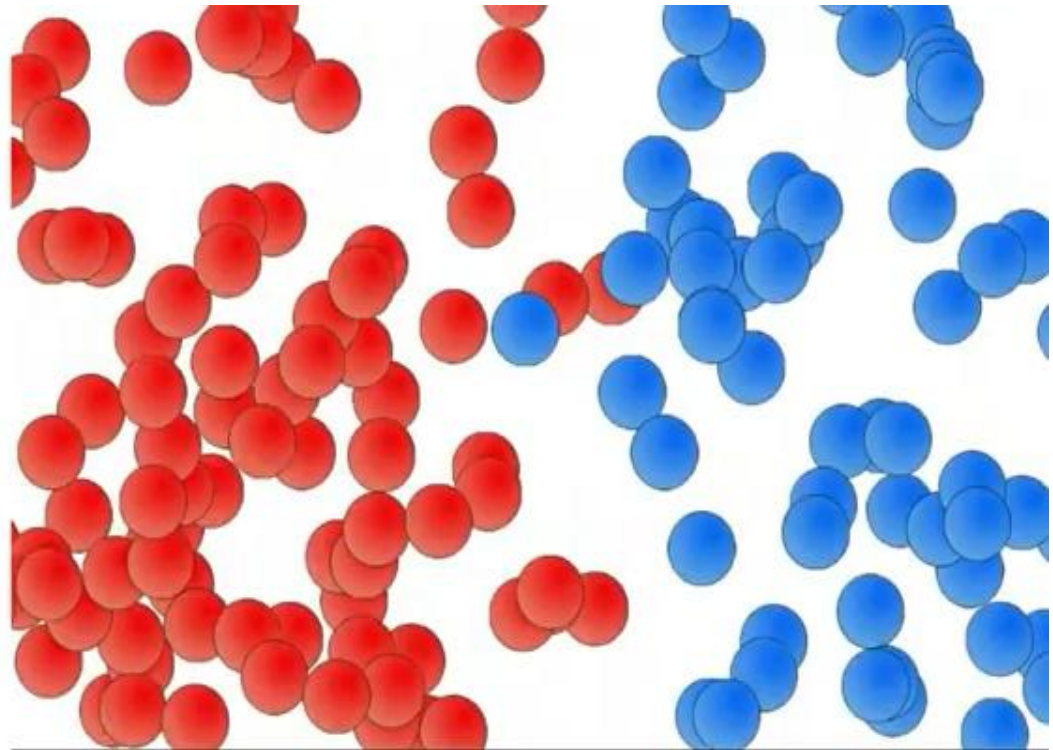
$$\frac{|\vec{v}_{n,p}|}{\vec{E}} = \mu_{n,p}$$

$$\frac{\left[\frac{cm}{s} \right]}{\left[\frac{V}{cm} \right]} = \left[\frac{cm^2}{V \cdot s} \right]$$

Current type 2: Diffusion current



Simulation of pollen (large particle) pushed around by atoms in a fluid

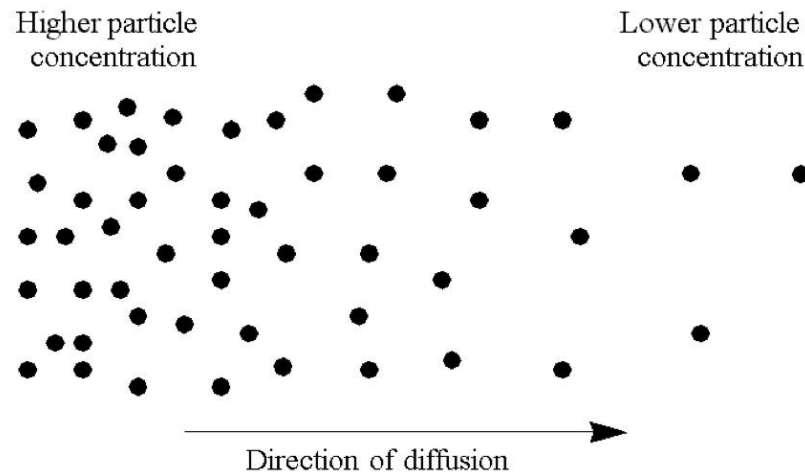


Current type 2: Diffusion current

- Remember Brownian motion of electrons & holes!
- When E-field = 0, but $T > 0$, thermal velocity $v_T = \sim 10^5 \text{ m/s}$
- But net drift velocity $v_d = 0$
- So net current $J_d = 0$

Brownian motion:
"A random walk"

- What if there is a *concentration or temperature* (thermal velocity) gradient?



Diffusion in semiconductors

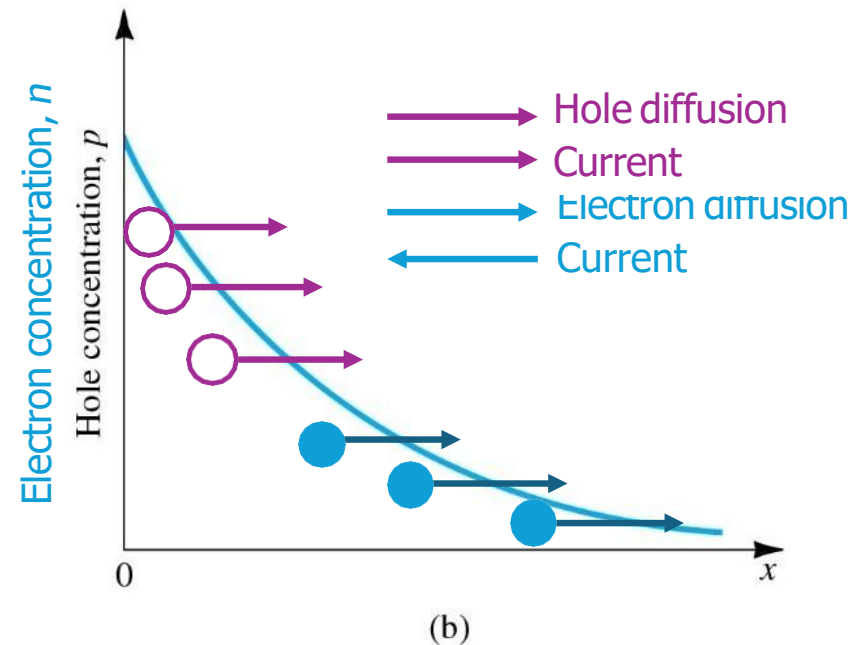
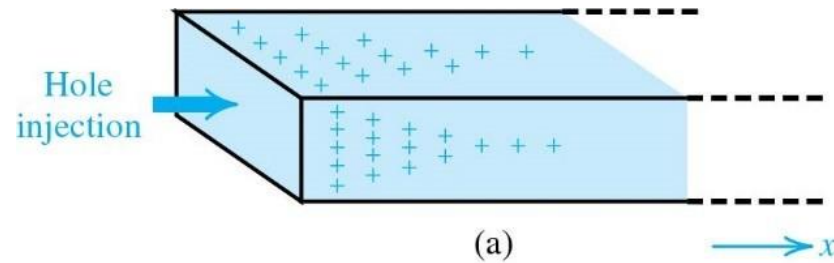
Starts with a difference in concentration of carriers across the material...

Current densities defined by:

$$J_{D,p} = -qD_p \frac{dp(x)}{dx} \left[\frac{A}{cm^2} \right]$$

$$J_{D,n} = qD_n \frac{dn(x)}{dx} \left[\frac{A}{cm^2} \right]$$

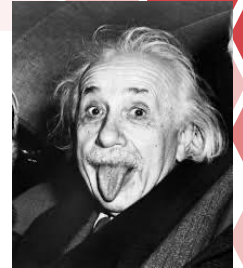
D_n, D_p are the diffusivities (diffusion constants) of electrons, holes. Typical values are $D_n = 35 \frac{cm^2}{s}$, $D_p = 12 \frac{cm^2}{s}$



Einstein relationship

$$\frac{D}{\mu} = \frac{k_B T}{q} = V_T$$

<< This is very important!



$k_B = 8.617 \times 10^{-5} \text{ eV/K}$
or
 $k_B = 1.38 \times 10^{-23} \text{ J/K}$

it connects diffusivity with mobility,
which we already know how to look up.
(In other words, it describes how quickly
a particle will move in a medium due to a
concentration gradient.)

T = temperature [K]

V_T = Thermal voltage $\sim 25.9 \text{ mV}$ at $T = 300 \text{ K}$

***pn*-junction (no applied voltage)**

doped: p-type



in equilibrium

"Quasi-neutral" = For the scales we're dealing with, the **semiconductor is almost electrically neutral.**

$$p = N_A$$

(acceptors are slightly negative)

Fixed atoms
Mobile charge carriers

doped: n-type

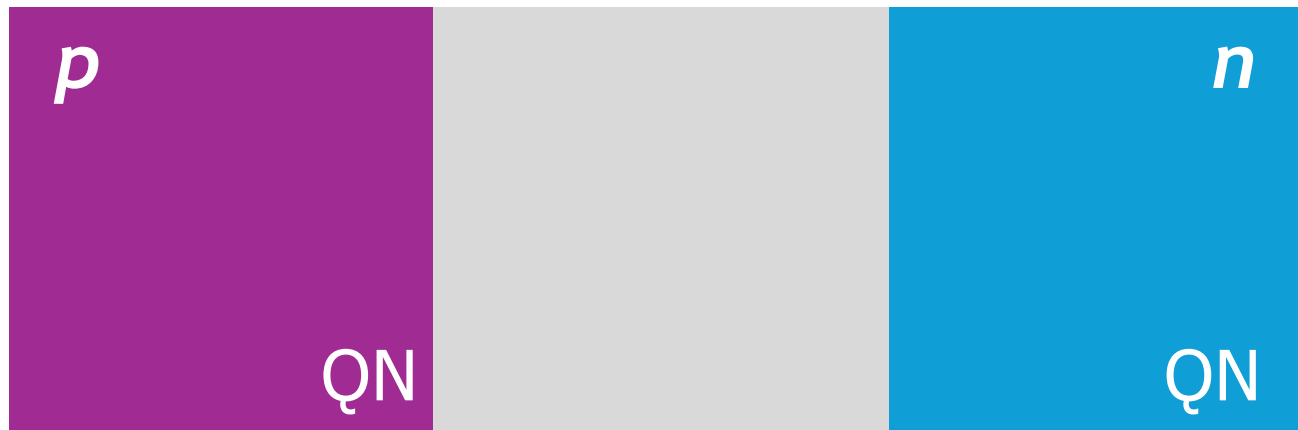
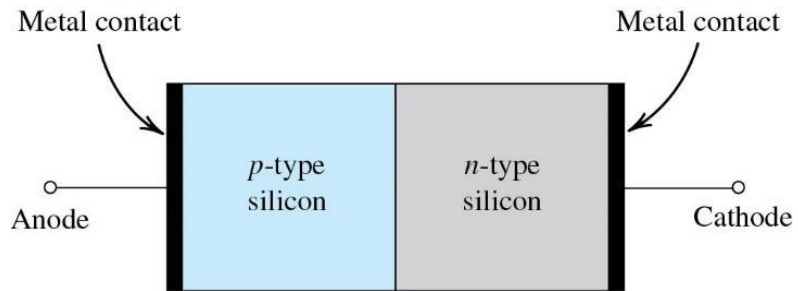


$$n = N_D$$

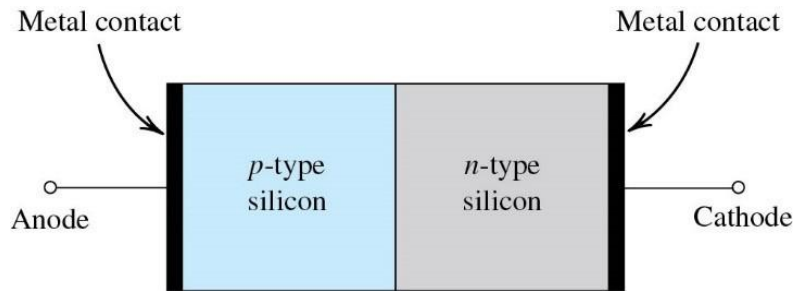
(donors are slightly positive)

Fixed atoms
Mobile charge carriers

***pn*-junction (no applied voltage)**

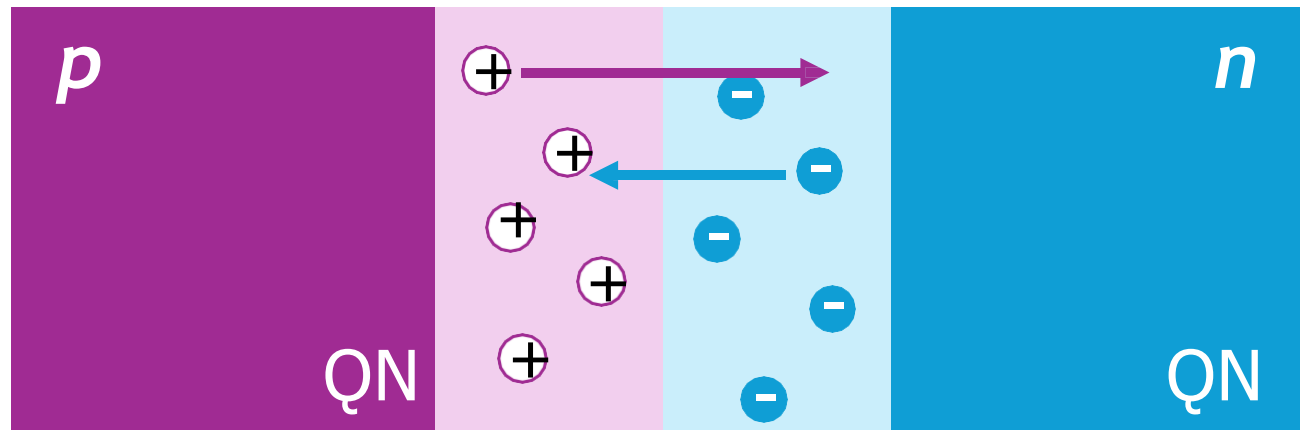


pn-junction (no applied voltage)



The holes recombine with some **majority electron carriers** on the **right**, causing a *depletion of free electrons* close to the junction

The electrons recombine with some **majority hole carriers** on the **left**, causing a *depletion of holes* close to the junction



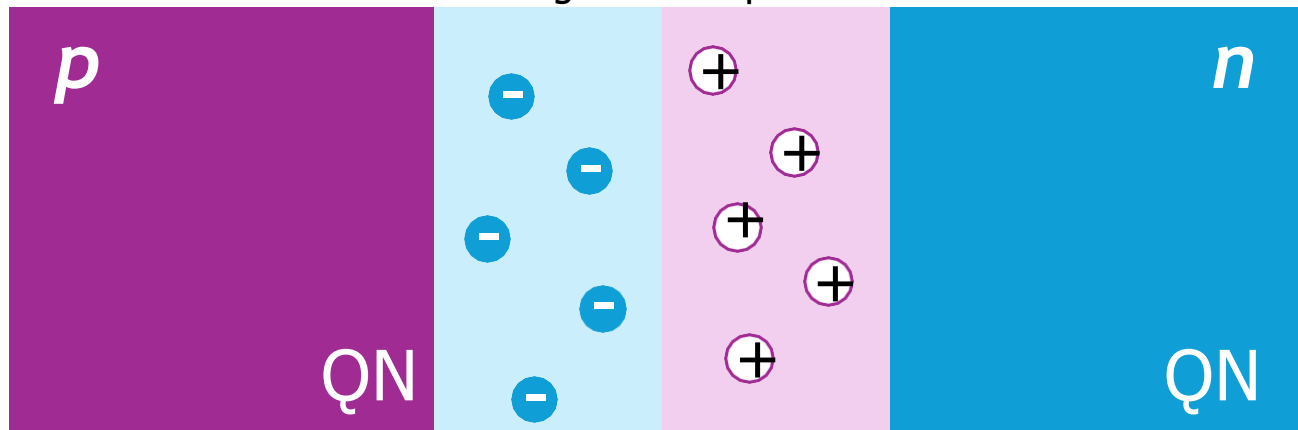
$I_D = \text{diffusion current}$

***pn*-junction (no applied voltage)**

Net charge in
depletion region is
 $q(N_D - N_A)$

Only made up of fixed
charge sites (N_A and N_D)

There are NO mobile charges
in the depletion region

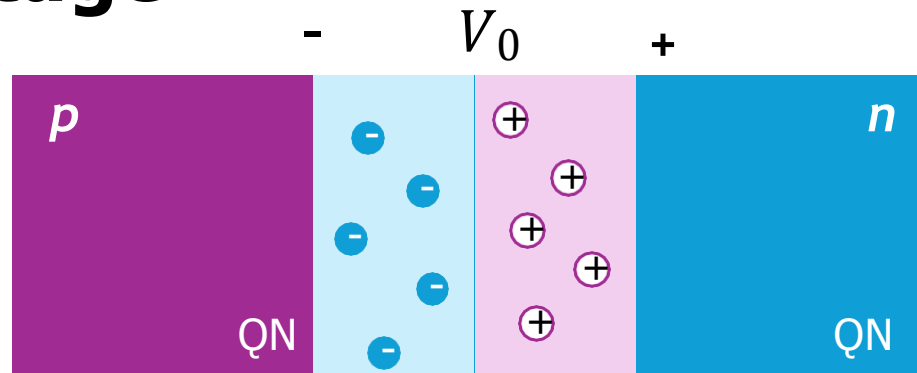


The electric field opposes the
diffusion of more holes into n region
and more electrons into p region

A voltage potential is established!!!

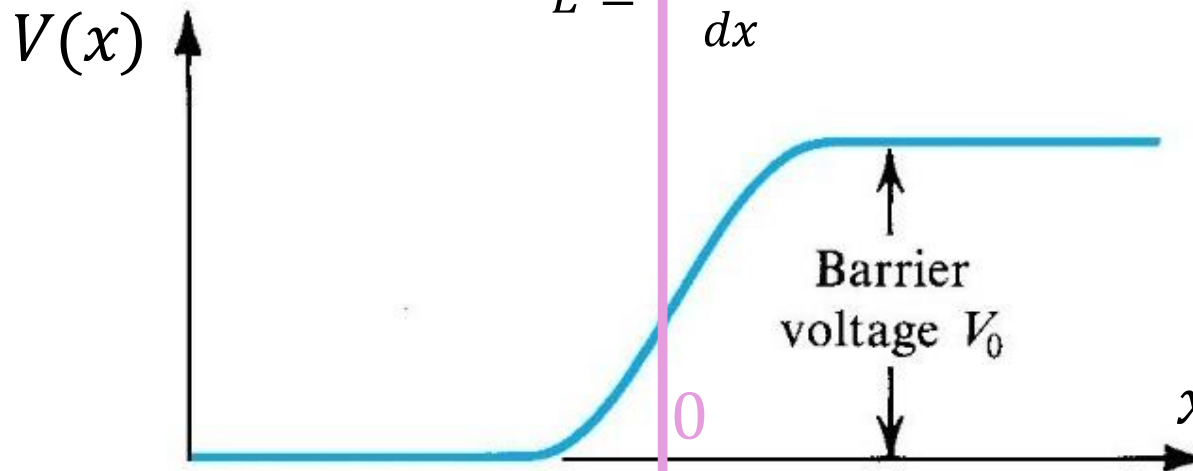
Built-in voltage

V_0 CANNOT be measured as an open-circuit voltage!
(opposing charge carriers at the contacts counteract this)



Electric field

$$E = -\frac{dV}{dx}$$



Voltage potential

With no voltage applied, barrier voltage is:

$$V_0 = V_T \ln \left(\frac{N_A N_D}{(n_i)^2} \right)$$

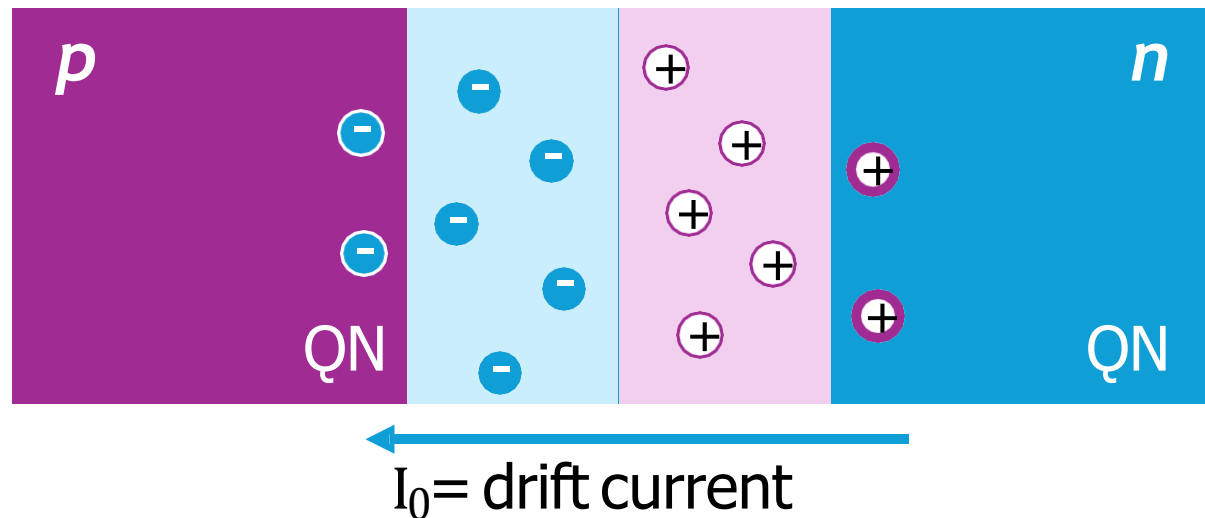
N_A = doping concentration in p-region

N_D = doping concentration in n-region

V_0 typ 0.6 – 0.9 V at room temp

Minority carrier drift

Once the electric field is established, some of the thermally-generated carriers drift towards the junction C get swept away, into the opposite side of the depletion region.



The total drift current, I_S , is the sum of:

- (1) the drift of thermally-generated holes in the n-region into the p-region
- (2) the drift of thermally-generated electrons in the p-region into the n-region

Under open-circuit conditions, the magnitudes of diffusion current = drift current.

$$I_D = I_S$$

Total currents in a semiconductor

Recall: In equilibrium (open circuit conditions) total current must be zero (KCL)

$$J_{\text{drift}} = -J_{\text{diffusion}}$$

or, for electrons: $J_{n,\text{drift}} + J_{n,\text{diffusion}} = 0$

- So any disturbance (e.g. light, doping gradient, thermal gradient) which may set up a carrier concentration gradient, will also internally set up a built-in current

Total currents in a semiconductor

- Now, we can FINALLY write down the TOTAL currents...

- For electrons: $J_N = J_{N,\text{drift}} + J_{N,\text{diff}} = qn\mu_n\mathcal{E} + qD_N \frac{dn}{dx}$

- For holes: $J_P = J_{P,\text{drift}} + J_{P,\text{diff}} = qp\mu_p\mathcal{E} - qD_P \frac{dp}{dx}$

- And TOTAL current:

$$J_{\text{TOT}} = J_N + J_P$$

**minority carriers contribute little to drift current (usually, too few of them!), BUT if their gradient is high enough...